

Determinants of Diabetes Mellitus in the Pima Indians

WILLIAM C. KNOWLER, MD, DRPH
MOHAMMED F. SAAD, MD, MRCP
DAVID J. PETTIT, MD

ROBERT G. NELSON, MD, MPH
PETER H. BENNETT, MB, FRCP, FFCM

OBJECTIVE— To review the research findings on the determinants of diabetes mellitus in Pima Indians.

RESEARCH DESIGN AND METHODS— Pima Indians in Arizona have participated in a longitudinal diabetes study that has provided data on and hypotheses about the development of NIDDM. Findings from this study are reviewed and updated.

RESULTS— Frequency distributions of plasma glucose and HbA_{1c} are bimodal in Pima adults, and substantial risk of the specific vascular complications of diabetes is confined to those in the higher components of these distributions. These findings contributed to the adoption of internationally recognized criteria for classification of glucose tolerance. Diabetes in the Pimas is strongly familial, and probably of genetic origin, although the precise nature of the gene or genes involved remains unknown. Obesity, which is at least in part environmentally determined, is a major factor interacting with the presumed genetic susceptibility to result in diabetes. The incidence of diabetes in the Pimas has increased during the last several decades, providing further evidence for environmental-genetic interaction. Longitudinal studies suggest that the progression from normal to diabetes can be considered to involve two stages. The first, primarily attributable to insulin resistance, leads to impaired glucose tolerance, and the second, which depends on insulin secretory failure, leads to worsening hyperglycemia and overt diabetes.

CONCLUSIONS— The Pimas and many other American Indian populations suffer from a high incidence of diabetes and its characteristic disabling or fatal complications, and would benefit from continued research on the pathogenesis and prevention of the disease.

N IDDM is a very common and serious disease in many American Indian tribes (1). In this study, the epidemiology of diabetes is described in the Pima Indians of Arizona, an American Indian tribe with the highest re-

corded prevalence and incidence of NIDDM of any geographically defined population (2).

The Pima Indians and the closely related Tohono O'odham (Papago) Indians who live in a part of the Gila River

Indian Community in the desert of central Arizona have participated in a comprehensive longitudinal study of the epidemiology of diabetes mellitus, for which they have been examined at intervals of ~2 yr since 1965 (3). This study provides much of the information on the prevalence, incidence, risk factors, and pathogenesis of diabetes in the Pima Indian population, discussed in this and previous reviews (4,5).

BACKGROUND OF STUDY— Ancestors of the present-day Pima Indians are believed to have lived in the valleys of the Gila and Salt Rivers, in what is now Arizona, for >2000 yr. According to Haury (6), the Pimas descended from the Hohokam, who represented one of the groups of Paleoindians, who originated from the first of three migrations across the Bering land bridge from Asia to North America. Evidence for the Pimas' descent from Paleoindians rather than from people of later migrations from Asia comes from comparisons of culture and language (7), blood groups (7), dental morphology (8), serum immunoglobulin allotypes (9), and variations in mitochondrial DNA (10).

The Hohokam and Pima Indians adapted successfully to the desert environment by growing irrigated crops, hunting, and gathering food. Their successful agriculture continued until the end of the 19th century, at which time settlement of the area by people of European derivation and diversion of the water supply disrupted the Pimas' agriculture (11), which has gradually been reestablished with water from the reservoir of the Coolidge Dam completed in 1929 on the Gila River (12). Although agriculture is now the major tribal industry, crops are sold or fed to livestock; thus, the Pimas purchase rather than produce most of their food.

The Pimas now suffer from high prevalence rates of NIDDM and obesity. The high prevalence of diabetes appears to be a recent phenomenon, although

FROM THE DIABETES AND ARTHRITIS EPIDEMIOLOGY SECTION, PHOENIX EPIDEMIOLOGY AND CLINICAL RESEARCH BRANCH, NATIONAL INSTITUTE OF DIABETES AND DIGESTIVE AND KIDNEY DISEASES; AND THE CLEVELAND CLINIC FOUNDATION, PHOENIX, ARIZONA.

ADDRESS CORRESPONDENCE TO WILLIAM C. KNOWLER, MD, DRPH, DIABETES AND ARTHRITIS EPIDEMIOLOGY SECTION, PHOENIX EPIDEMIOLOGY AND CLINICAL RESEARCH BRANCH, NATIONAL INSTITUTE OF DIABETES AND DIGESTIVE AND KIDNEY DISEASES, 1550 EAST INDIAN SCHOOL ROAD, PHOENIX, AZ 85014.

NIDDM, NON-INSULIN-DEPENDENT DIABETES MELLITUS; BMI, BODY MASS INDEX; WHO, WORLD HEALTH ORGANIZATION; NDDG, NATIONAL DIABETES DATA GROUP.

temporal changes in the prevalence of diabetes are difficult to estimate because of changing diagnostic methods. In the early part of this century, Hrdlička (13) reported only one case of diabetes in the population, and Russell (11) listed 28 diseases affecting the Pimas without mentioning diabetes. In 1940, Joslin (14), based on a review of medical records, reported that the prevalence of diabetes in American Indians was similar to that of the general population. Diabetes has been reported as being excessively common in the Pimas since the 1950s (15,16). In the mid-1960s, it had become a major health problem (3), and its prevalence has increased since that time (17), affecting, by 1981, ~50% of the adults who are ≥ 35 yr of age (18).

Pima Indians with diabetes of recent onset do not have subnormal insulin or C-peptide concentrations. Except in the presence of a complicating illness, diabetic Pimas do not require exogenous insulin to prevent ketosis, even when the disease develops at young ages (19,20). High titers of islet cell antibodies are not found (21). The Pimas therefore have NIDDM.

RESEARCH DESIGN AND

METHODS — Since 1965, a longitudinal study of diabetes and its complications has been conducted among the residents (mostly Pima Indians) of a part of the Gila River Indian Community. Approximately every 2 yr, each resident ≥ 5 yr of age is invited for a biennial examination that includes measurements of height and weight, a medical history, and a glucose tolerance test in which glucose concentration is measured in venous plasma obtained fasting and 2 h after ingestion of 75 g glucose-equivalent carbohydrate. Serum insulin concentrations also have been measured since 1975. Because examinations are conducted periodically, the incidence, or rate of development of new cases of diabetes, can be determined, facilitating study of factors that predict diabetes. The same methods are used for subjects of all ages, and data on family relationships are recorded. Thus, the family data presented

are based on the results of examinations rather than on family history.

DIAGNOSTIC CRITERIA — Because diabetes is defined by hyperglycemia, the frequency distributions of fasting and 2-h postload plasma glucose concentrations and of HbA_{1c}, an integrated measure of glycemia during the preceding weeks were examined and are shown in Fig. 1 for subjects ≥ 35 yr of age (5). These distributions are bimodal and conform to a model of two overlapping normal distributions. These results suggest that diabetes, as defined by glucose or HbA_{1c} concentrations in the upper component of these distributions, is a distinct entity rather than simply representing the tail of a skewed distribution. At young ages nearly all subjects are in the lower component of each distribution, and the proportion of people belonging to the upper component increases with age.

Prevalence rates of other manifestations of diabetes, such as diabetic retinopathy and proteinuria, are much greater in the higher component of each distribution (5,24–28). Thus, people in the upper component of each distribution have a distinct syndrome characterized by hyperglycemia and a high risk of microvascular complications.

Such findings from the Pima Indians and from other studies (29,30) led to the adoption of new standardized criteria for the diagnosis of diabetes by the NDDG (31) and the WHO (32,33). These levels approximate the separation of the two components of the distributions of glucose concentrations shown in Fig. 1.

The WHO and the NDDG also define impaired glucose tolerance by postload glucose concentrations above normal but below those diagnostic of diabetes, although their criteria for this condition differ (34). Impaired glucose tolerance is not generally accompanied by symptoms or microvascular complications characteristic of diabetes, but is associated with increased risk of developing diabetes. In the Pima Indian longitudinal study, diabetes and impaired

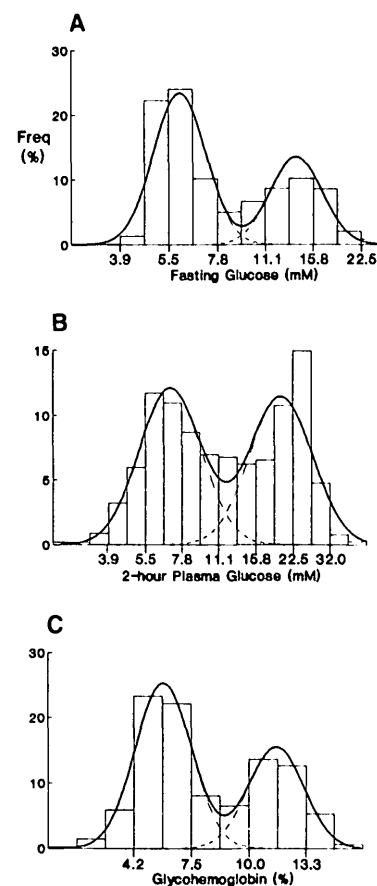


Figure 1—Frequency distributions of fasting plasma glucose concentrations (A), 2-h plasma glucose concentrations (B), and HbA_{1c} (C). Subjects ≥ 35 yr old ($n = 901$) are included regardless of previously diagnosed diabetes or hypoglycemic treatment. HbA_{1c} was measured by electrophoresis (22). The bars show frequencies on a logarithmic scale. Each curve represents a mathematical model of two overlapping normal distributions with a common variance fit by maximum likelihood (23). Adapted from Knowler et al. (5).

glucose tolerance have been classified according to WHO criteria (2; Table 1).

PREVALENCE AND INCIDENCE OF DIABETES

Prevalence

Table 2 shows the prevalence of diabetes during three time periods from 1965 to 1990. Prevalence increases with age and

Table 1—WHO criteria for the oral glucose tolerance test

	VENOUS PLASMA GLUCOSE (mM)*		
	FASTING		2-H
NORMAL GLUCOSE TOLERANCE†	<7.8	AND	<7.8
IMPAIRED GLUCOSE TOLERANCE	<7.8	AND	7.8–11.0
DIABETES	≥7.8	OR	≥11.1

*Cut points for serum glucose or capillary whole blood are slightly different and can be found in ref. 33 (7.8 mM [140 mg/dl], 11.1 mM [200 mg/dl]).

†Although the term normal is not used by WHO (33), we use it here to mean the absence of impaired glucose tolerance and diabetes.

tends to reach a plateau in older ages. In most age and sex groups, the prevalence of diabetes is higher in the more recent periods than in 1965–1974. Thus, the historical evidence previously cited that the prevalence of diabetes has increased during this century is extended by direct observations during the last 25 yr. The prevalence may show an increase because of an increase in incidence (the rate at which new cases develop) or because of greater improvement in survival of diabetic than of nondiabetic people in the latter time periods, although diabetes

has little impact on the overall mortality rates <55 yr of age (35).

Incidence

Incidence is the rate (per time) at which new cases of disease develop among people initially without the disease. From the beginning of the study in 1965 until 1990, there were 736 new cases of diabetes among >41,000 person-yr of follow-up of people nondiabetic at first examination. As shown in Fig. 2, these rates have increased in most age groups during the 25 yr of the study (36). The

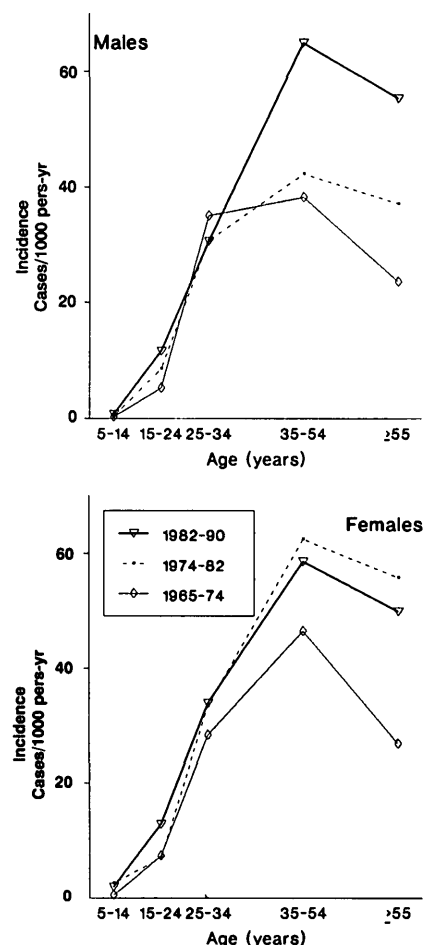


Figure 2—Age- and sex-specific incidence rates of diabetes in Pima Indians during three time periods of this study. Subjects were of at least half Pima or Tohono O'odham heritage and residents of the study area. Incidence rates are expressed as new cases of diabetes/1000 person-yr observation of nondiabetic subjects. Age- and sex-specific incidence rates were computed for the first three 8- or 9-yr time periods of the study, February 1965–January 1974, February 1974–January 1982, and February 1982–January 1990. In total, 736 cases of diabetes developed during 41,844 person-yr of follow-up. Adapted from Knowler et al. (36).

Table 2—Prevalence (%) of diabetes in Pima Indians by age, sex, and time period

		PERIOD OF STUDY		
		1965–1974	1974–1982	1982–1990
MALES	5–14	0	0	0
	15–24	2	5	5
	25–34	22	24	20
	35–44	37	53	49
	45–54	45	50	67
	55–64	42	55	69
	≥65	39	59	65
FEMALES	5–14	0	1	1
	15–24	1	4	5
	25–34	22	22	21
	35–44	39	50	46
	45–54	53	67	62
	55–64	71	72	86
	≥65	58	71	74

For each period data from each subject obtained at the biennial examination closest to the midpoint of the period are used.

rates appear to be rising still in men ≥35 yr of age but to have stabilized in the women. In all time periods in both sexes the incidence rates are lower in people ≥55 yr of age than in younger adults (2).

The increasing incidence rates of

diabetes preceding and during this study provide evidence for environmental risk factors for diabetes, because gene frequencies cannot change fast enough to account for these changes in incidence. The disease has also apparently increased in frequency in other populations (37–42), including many American Indian tribes (43) that have experienced major socioeconomic changes or migration, often accompanied by increasing obesity and decreasing physical activity.

RISK FACTORS FOR DIABETES —

Several risk factors predicting the development of diabetes have been examined in Pima Indians. These include parental diabetes, genetic markers, obesity, and diet.

Parental diabetes

The age-specific prevalence of diabetes in offspring is shown in Fig. 3 according to parental diabetes among subjects whose parents have been examined (36). Offspring are classified into three groups according to diabetes in their parents: 1) in the early group ($n = 1397$), at least one parent developed diabetes <45 yr of age; 2) in the late group ($n = 528$), at least one parent developed diabetes >45 yr of age, but neither had diabetes before this age; and 3) in the nondiabetic group ($n = 199$), both parents were nondiabetic until ≥ 45 yr of age. Parental data for offspring ≥ 45 yr old are too sparse for inclusion in this analysis.

Prevalence rates are highest in offspring whose parents developed diabetes at an early age. Below 25 yr of age, diabetes occurs only in people with at least one diabetic parent and is much more frequent if that parent had diabetes of early onset. This observation and examination of pedigrees (44,45) suggest an autosomal dominant mode of inheritance. The prevalence of diabetes in older people, however, is high, even among the offspring of two nondiabetic parents, and thus is not consistent with transmission by a dominant gene with high penetrance.

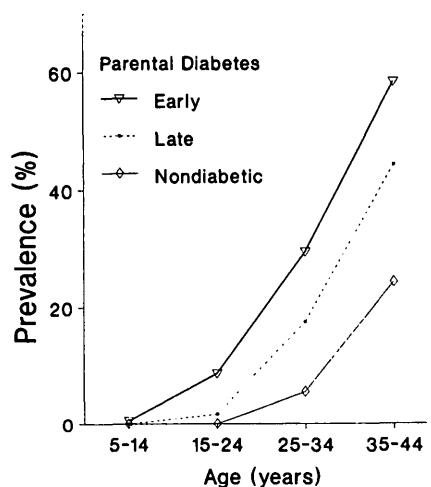


Figure 3—Prevalence of diabetes according to presence and age of onset of diabetes in parents. People for whom both parents had been examined were grouped into three parental diabetes groups: early, late, or nondiabetic according to whether the parents had diabetes and whether the parental diabetes was diagnosed <45 or >45 yr of age, as defined in the text. Adapted from Knowler et al. (36).

The interpretation of transmission of susceptibility to diabetes from parent to offspring is complicated by a higher risk of obesity and diabetes in offspring exposed to diabetes in utero (46–49). This finding is discussed in detail elsewhere (this issue, Pettitt et al., p. 310–14).

Admixture and genetic markers

Clues to the location of disease-susceptibility genes have often been sought by examining associations between a disease and a genetic marker (i.e., a polymorphic characteristic determined by a known gene). The implications of such associations have been illustrated in the Pima Indian study with the haplotype $Gm^{3:5,13,14}$ from the Gm system of IgG (50). The presence of this haplotype is inversely associated with diabetes in each age-group ≥ 15 yr, as shown in Fig. 4. In each age-group, the prevalence of diabetes in subjects with $Gm^{3:5,13,14}$ is only $\sim 50\%$ that of subjects

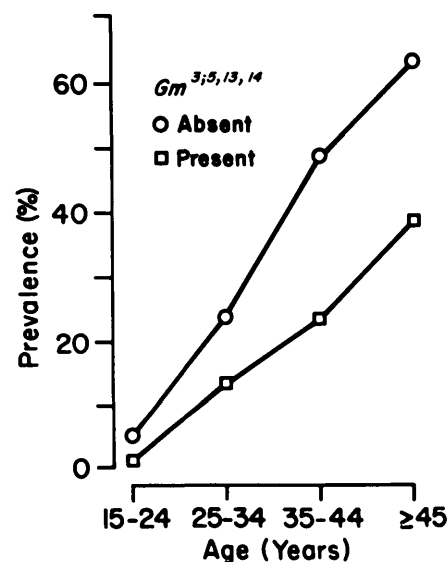


Figure 4—Prevalence of diabetes by age and presence of the haplotype $Gm^{3:5,13,14}$ among 4095 residents of the Gila River Indian Community. From Knowler et al. (50). Reprinted with permission from the American Journal of Human Genetics.

without this haplotype, suggesting that this haplotype (or a gene in linkage disequilibrium) confers protection from diabetes.

A different explanation for the association is provided, however, by considering the degree of Indian heritage of each subject. The vast majority are of full Indian heritage, but some have different degrees of non-Indian (predominantly caucasian) admixture. Age-adjusted prevalence rates of diabetes are shown in Fig. 5 according to amount of Indian heritage and presence of the $Gm^{3:5,13,14}$ haplotype. Diabetes prevalence is directly related to Indian heritage, but within heritage classes, is unrelated to Gm type. Thus, the primary association is between diabetes and Indian heritage, and the association of diabetes with Gm type is attributable to the relationship of each of these factors to Indian heritage, as shown in Fig. 6. Compared with full-heritage Indians, people with caucasian admixture are more likely to have the

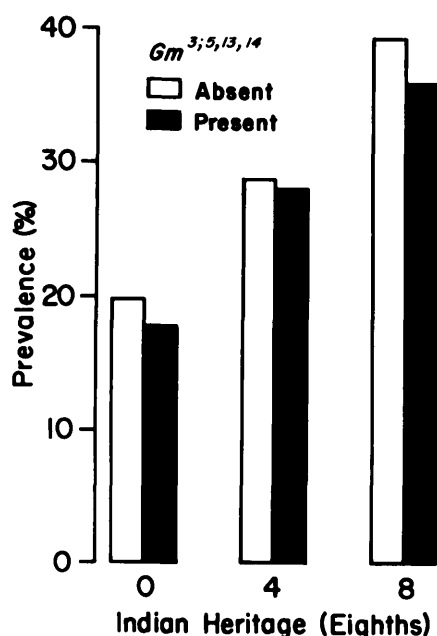


Figure 5—Age-adjusted prevalence of diabetes by presence of the haplotype $Gm^{3;5,13,14}$, according to Indian heritage, among residents of the Gila River Indian Community. Prevalence rates were age-adjusted in people ≥ 15 yr (32, 217, and 3656 were of 0, 4, or 8 eighths Indian heritage). From Knowler et al. (50). Reprinted with permission from the American Journal of Human Genetics.

$Gm^{3;5,13,14}$ haplotype but less likely to have diabetes. Therefore, despite an association between diabetes and this genetic marker, it appears that the marker plays no direct role in disease susceptibility. The strong relationship of diabetes to Indian heritage (Figs. 5 and 6), however, is evidence for a genetic basis to the disease, but does not indicate the number or location of such genes.

Diabetes in Pima Indians is also associated with the HLA-A2 phenotype (51). Unlike the association with Gm type, the association of diabetes with HLA-A2 is found among people reporting full Pima Indian heritage. Thus, it cannot be explained by non-Indian admixture, but might result from linkage disequilibrium with a disease-susceptibility locus. Diabetes is not associated

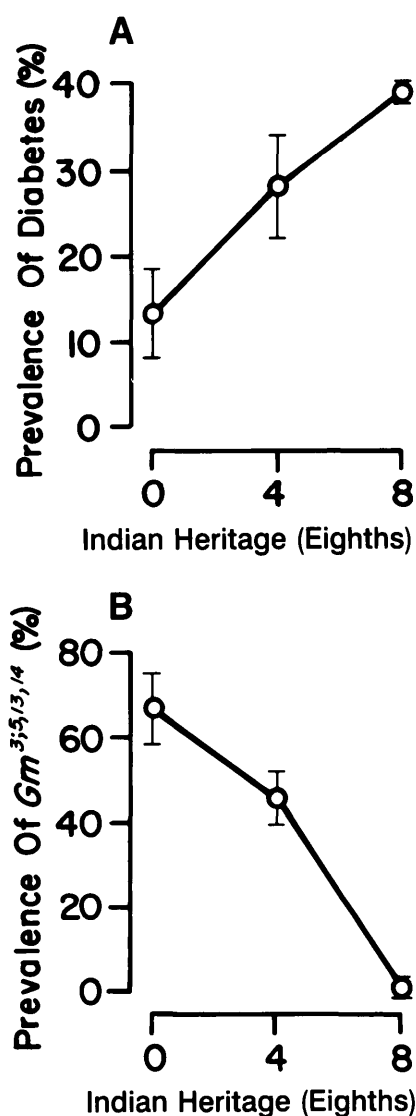


Figure 6—Age-adjusted prevalence (± 1 SE) of diabetes (A) and of $Gm^{3;5,13,14}$ (B), according to Indian heritage, among 3905 residents of the Gila River Indian Community. From Knowler et al. (50). Reprinted with permission from the American Journal of Human Genetics.

with polymorphic alleles at the insulin (52) and insulin receptor (53) genes in Pima Indians. Systematic studies of Pima Indian families for linkage of diabetes with other genetic markers are being performed (54). Linkage studies will also be performed for measures of in vivo insulin action, a risk factor for diabetes, for

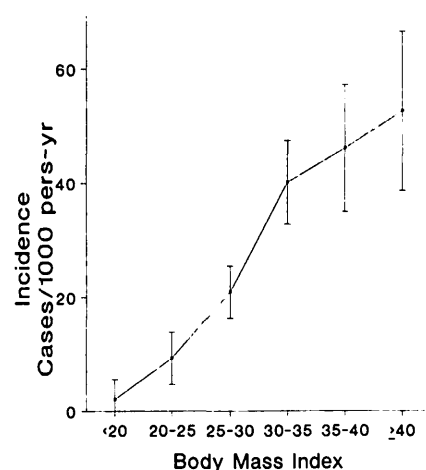


Figure 7—Age- and sex-adjusted incidence of diabetes by BMI, with 95% confidence intervals. The BMI was determined at each biennial examination and assessed as a predictor of development of diabetes by the next examination, which must have occurred within 5 yr. There were 542 new cases of diabetes during 29,710 person-yr of follow-up. Incidence rates are adjusted by age and sex, using the 1980 U.S. white population as a standard, to give average incidence rates according to BMI. Adapted from Knowler et al. (18).

which a trimodal frequency distribution suggests a major gene effect at a single locus (55).

Obesity

The association of NIDDM with obesity is well known. The age- and sex-adjusted incidence rates of diabetes in the Pimas are shown in Fig. 7 as a function of the BMI (kg/m^2) measured in nondiabetic subjects (18). Although there is a marked effect of obesity, even in Pima Indians with BMIs between 20 and 25 kg/m^2 , which are below the average for the U.S. population in most age-groups, the age- and sex-adjusted incidence rate of diabetes is about eight times that of the predominantly caucasian population of Rochester, MN (18). Thus, obesity magnifies the risk of diabetes, which is high even in the nonobese Indians. Although there has been some increase in

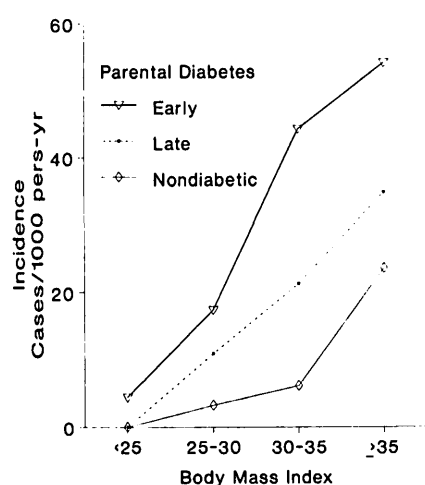


Figure 8—Age- and sex-adjusted incidence of diabetes by BMI and parental diabetes in people aged 15–34 yr. Adjusted incidence rates are shown for subjects in three groups according to parental diabetes, as defined in Fig. 3. Diabetes developed among 197 of these subjects during 11,288 person-yr of follow-up. Adapted from Knowler et al. (18) and (36).

obesity during the period of this study (5), it is not sufficient to account for the increasing incidence rates of diabetes (36).

Obesity interacts with parental diabetes, another strong diabetes risk factor. Age-adjusted diabetes incidence rates are shown in Fig. 8 by BMI and parental diabetes (18,36). Obesity and parental diabetes increase diabetes incidence rates synergistically, with the highest rates seen in obese subjects with at least one parent with diabetes of early onset. The incidence of diabetes is much lower in thin Pima Indians, even among those with diabetic parents.

Diet

Although the Pima Indian diet has undergone tremendous change during the last century as a result of disruption of the traditional agricultural life-style by European-derived settlers, as reviewed elsewhere (5,56), it is difficult to determine the effect this has had on the emergence of diabetes. According to a dietary

study of Pima Indian women in 1969, the average Pima diet was similar to the average U.S. diet, as estimated from food purchase data. Pima Indian women 25–44 yr of age consumed an average 3163 kcal/day, of which 44% were from carbohydrate, 12% from protein, and 44% from fat. There was no significant difference found in the calorie intake of those with and without diabetes, although the diabetic women reported lower sugar consumption (56).

Among 187 of the women who were nondiabetic at this survey and were followed during the next decade, 87 developed diabetes. The incidence of diabetes was positively related to total calorie and carbohydrate intake at the time of the survey, but not to sucrose intake. Although carbohydrate intake was the single strongest predictor of the development of diabetes, total carbohydrate was strongly related to total calorie intake and fat consumption (57).

PATHOGENESIS OF DIABETES

Worsening glucose tolerance

If tested early in life, presumably all people would fall into the normal components of the bimodal frequency distributions of plasma glucose and HbA_{1c} (Fig. 1). Thus, the existence of a second component later in life implies that many people make an abrupt transition to the upper component and that the antimodal parts of the distributions (representing impaired glucose tolerance and the lower ranges of diabetic glucose concentrations) are relatively unstable. The stability has been examined in terms of the incidence of diabetes by 2-h postload plasma glucose concentrations. Age- and sex-adjusted incidence rates of diabetes are shown in Fig. 9 according to the 2-h postload plasma glucose at the preceding biennial examination (5,58). There is little relationship with 2-h glucose concentrations in the range considered normal by WHO criteria (<7.8 mM), but people with higher 2-h glucose concentrations (impaired glucose tolerance) have mark-

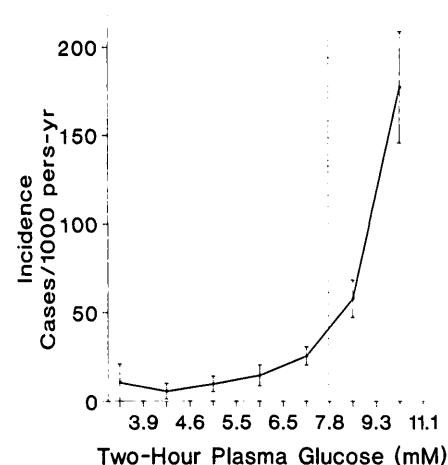


Figure 9—Age- and sex-adjusted incidence of diabetes by 2-h postload plasma glucose concentration, in subjects ≥ 15 yr. Incidence in new cases/1000 person-yr at risk is shown with 95% confidence intervals. The 2-h postload plasma glucose concentration was determined at each biennial examination and assessed as a predictor of development of diabetes by the next examination, which must have occurred within 5 yr. The vertical line at plasma glucose concentration equals 7.8 mM and represents the lower limit of impaired glucose tolerance. The rates are based on 542 new cases in 29,710 person-yr of follow-up. Adapted from Knowler et al. (5) and Bennett et al. (58).

edly higher incidence rates of diabetes (58–60) similar to findings in other populations (61–66).

Relationship of glucose and insulin concentrations

People with impaired glucose tolerance have not only higher postload glucose concentrations, but also higher fasting and postload serum insulin concentrations, on average, than those with normal glucose tolerance (58,60). The higher serum insulin concentrations compensate for the insulin resistance characteristic of this condition (67,68). The 2-h postload plasma glucose concentration is not the only predictor of diabetes among nondiabetic Pima Indians. Fasting and 2-h plasma glucose and

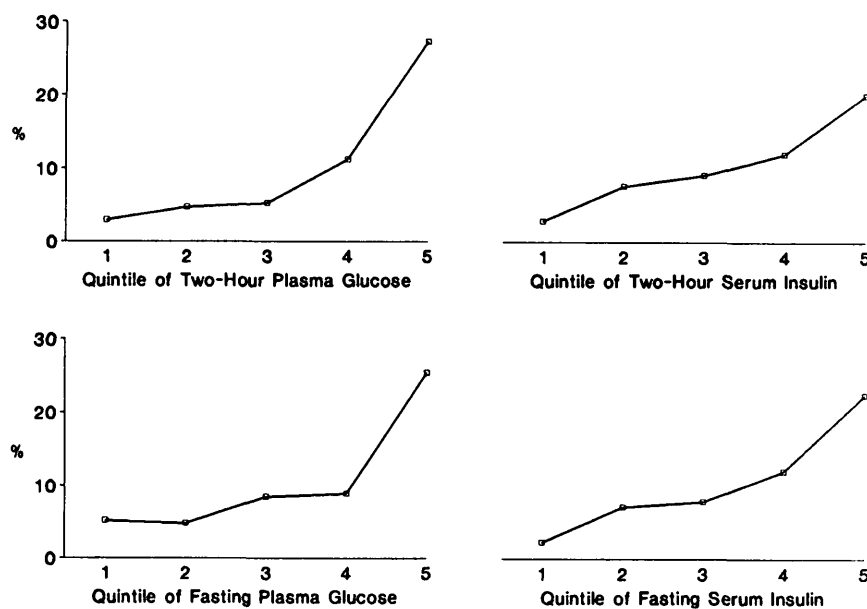


Figure 10—The 5-yr cumulative incidence of diabetes by quintiles of fasting and 2-h postload plasma glucose and serum insulin (69) concentrations. The cumulative incidence was estimated by the Kaplan-Meier method (70) in 1491 nondiabetic people ≥ 15 yr, with at least one follow-up examination after measurement of the variables. The quintiles were computed within decades of age, so the comparison between quintiles is not confounded by age. Adjusted for age with a proportional hazards analysis (71,72), the effect of each of these variables on the incidence of diabetes is highly significant ($P < 0.001$ for each variable). From Knowler et al. (5). Reprinted with permission from *Diabetes Metabolism Reviews*.

serum insulin concentrations each influence the incidence of diabetes, as shown in Fig. 10. The incidence shows a modest relationship with each of the glucose variables throughout the lower four quintiles, with a markedly increased rate among people in the highest quintile, whereas the relationships with insulin concentrations are more linear. Because the fasting and 2-h glucose and insulin concentrations among nondiabetic subjects are highly correlated with each other, they are examined in combination in Fig. 11. This figure shows that, among individuals with normal glucose tolerance, fasting serum insulin concentration is positively related to subsequent development of diabetes. Among those with impaired glucose tolerance, a higher fasting insulin concentration but a lower insulin response predict worsening to diabetes (60).

Cross-sectional studies of Pima Indians (52,58,73,74) and other groups (75–80) show that fasting and glucose-stimulated insulin concentrations form an inverted U-shaped curve when plotted against plasma glucose concentrations. Pima Indians follow the same pattern longitudinally as they develop diabetes, as shown in Fig. 12 (81). Fasting and 2-h insulin concentrations increase during the transition from normal to impaired glucose tolerance, probably reflecting the high and increasing insulin resistance that appears to be primarily responsible for early deterioration of glucose tolerance (68). The increase in insulin resistance may be attributable to weight gain (82–84), aging (85,86), and decreased physical activity (87). Deterioration of impaired glucose tolerance to diabetes is accompanied by a further increase in fasting insulin, but little change

in glucose-stimulated insulin concentrations. The transition to diabetes is probably caused not only by further increases in insulin resistance, as reflected by the increasing fasting insulin concentration, but also by failure to increase insulin secretion sufficiently in response to hyperglycemia. This finding is consistent with the observation that subjects with impaired glucose tolerance who have lower insulin responses to oral glucose are more likely to progress to diabetes (Fig. 11). After the onset of diabetes, both fasting and 2-h insulin concentrations diminish, despite further increases in plasma glucose concentrations. Thus, the development of NIDDM can be thought of as occurring in two stages: the first leading to impaired glucose tolerance with hyperinsulinemia (representing partially compensated insulin resistance) and the second leading to overt diabetes with worsening hyperglycemia as a result of failure to maintain sufficient compensatory hyperinsulinemia (perhaps as a result of β -cell defects or exhaustion) (88).

Origins of metabolic abnormalities that lead to diabetes

The familial transmission of diabetes in the Pimas suggests that susceptibility to diabetes with an early age of onset is inherited by the offspring. Evidence for genetic factors includes familial aggregation, not only of diabetes (18,36), but also of insulin-mediated glucose disposal rates (89) and severity of hyperglycemia (36). Genetic marker studies provide evidence for the importance of genetic factors but have not led to localization of a major gene. Insulin resistance is an early metabolic defect that may be a manifestation of a genetic defect causing susceptibility to diabetes (55,90).

Environmental factors also contribute to the development of diabetes. Obesity has a strong effect on the incidence of diabetes, although obesity itself may have genetic as well as environmental causes. The interaction between obesity and parental diabetes, and the in-

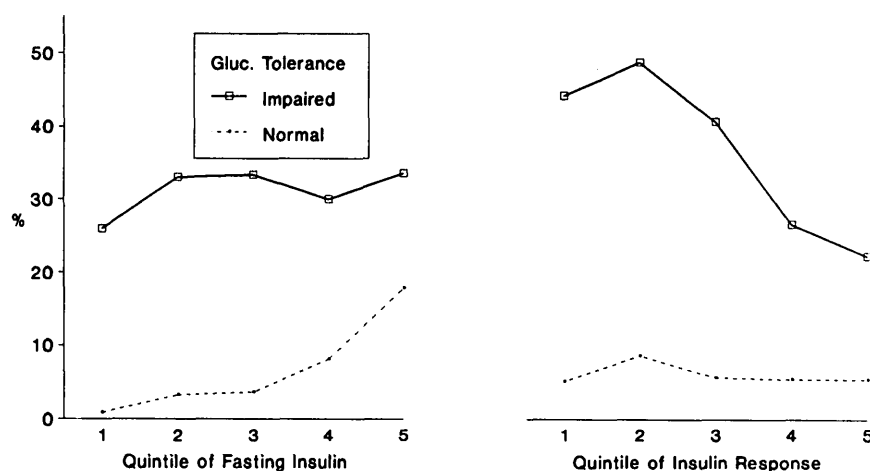


Figure 11—The 5-yr cumulative incidence of diabetes by quintiles of fasting insulin and insulin response to oral glucose in subjects with normal or impaired glucose tolerance, by WHO criteria (33). Incidence rates and quintiles of fasting insulin are defined as in Fig. 10. The insulin response is the ratio of the 2-h postload serum insulin concentration to the fasting insulin concentration. From Knowler et al. (5). Reprinted with permission from *Diabetes Metabolism Reviews*.

creasing incidence of diabetes over time, provide evidence for an interaction of genetic and environmental effects in the development of the disease.

OTHER AMERICAN INDIANS —

Although Pima Indians have been the most extensively studied American Indian population and have the highest recorded incidence of diabetes, many other unrelated Indian groups (such as the San Carlos Apache, Seminole, Choctaw, Cherokee, and Seneca Indians) also have unusually high frequencies of the disease (1). Diabetes appears to be more common in all Native American groups studied than in the caucasian population of the U.S.—except for Alaskan Eskimos (91). Diabetes is also unusually frequent among the Mexican-American population, whose gene pool represents a mixture of caucasian and American Indian genes (92). The predisposition of American Indians to diabetes may have arisen because diabetes-susceptibility genes were, by chance, very common among the original American Indians and their descendants, or because selective pressures favored survival of those who car-

ried the diabetes-susceptibility genotype. Only in the past century, however, has diabetes emerged as a major health problem. The increasing prevalence in the Pimas, together with circumstantial evidence of increasing frequencies in other tribes, suggest strongly that environmental factors have a major effect on the expression of the diabetes-susceptibility genotype.

CONCLUSIONS AND IMPLICATIONS FOR PREVENTION —

Genetic and environmental factors contribute to the high incidence of diabetes in the Pima Indians. Many of the same risk factors appear to play a role in other racial groups. Environmental factors (such as diet, physical activity, and perhaps other factors contributing to obesity) appear to have the most immediate relevance to prevention. The causes and mechanisms of the major physiological abnormalities leading to NIDDM, i.e., insulin resistance and inadequate insulin secretion, are not fully understood, but probably include both genetic and nongenetic factors. They may be subject to intervention,

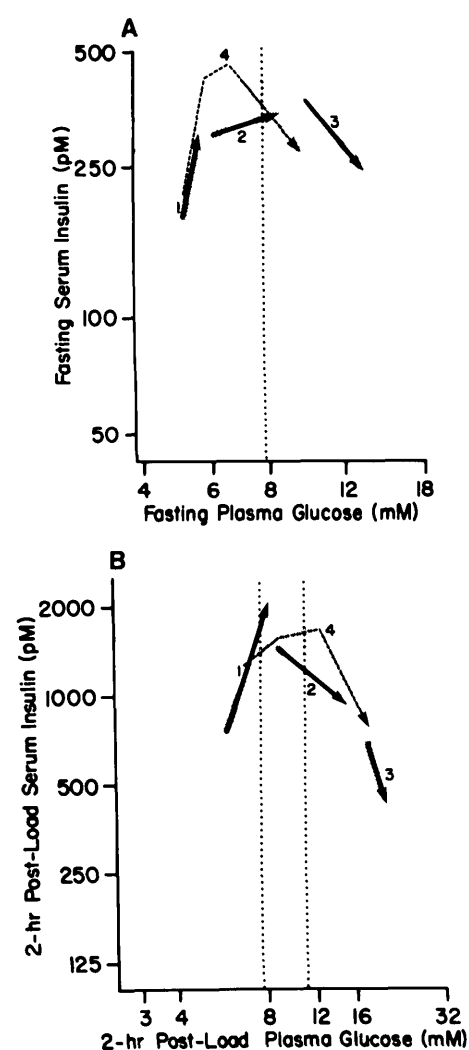


Figure 12—Changes in median fasting (A) and 2-h (B) serum insulin concentrations by corresponding plasma glucose concentrations in four groups of subjects. Subjects changed from normal to impaired glucose tolerance (group 1, $n = 81$), from impaired tolerance to diabetes (group 2, $n = 44$), and from the first to a subsequent examination with diabetes (group 3, $n = 27$). Subjects in group 4 ($n = 11$) were examined at each of these four stages. The shaded area shows the 95% confidence interval of the median serum insulin concentrations grouped by plasma glucose concentrations in 2817 Pima Indians aged ≥ 15 yr who were not taking hypoglycemic medicines. The vertical dotted lines represent the diagnostic threshold for diabetes based on fasting plasma glucose in the left panel, and the limits defining impaired glucose tolerance in the right panel. From Saad et al. (81). Reprinted with permission from *The Lancet*.

however, and studies of the effects of controlling obesity by diet and exercise, and of enhancing insulin secretion or action by drug treatment are needed, especially in view of the evidence that these treatments may prevent the deterioration of impaired glucose tolerance to diabetes (64).

Even if prevention is not possible, delaying the onset of diabetes would be extremely beneficial. The incidence of the chronic vascular complications responsible for most of the increased morbidity and mortality of diabetic persons is strongly related to the duration of diabetes (this issue, Nelson et al., p. 334–41). Thus, any delay in the onset of diabetes should reduce the impact of complications and increase longevity. Progress in research on the genetics of NIDDM should lead to means of identifying those people most likely to benefit from such intervention and perhaps also lead to better understanding of the nature of the underlying biochemical defect and the development of treatment specifically directed toward its correction.

Acknowledgments—We thank the members and leaders of the Gila River Indian Community, without whose participation and encouragement these studies could not have been performed; the Indian Health Service for continuing collaboration; and the staff of the Diabetes and Arthritis Epidemiology Section for conducting the studies.

References

1. Sievers ML, Fisher JR: Diabetes in North American Indians. In *Diabetes in America*. U.S. Department of Health and Human Services, Washington, DC, U.S. Govt. Printing Office, 1985, p. XI. 1–20 (NIH publ. no. 85–1468)
2. Knowler WC, Bennett PH, Hamman RF, Miller M: Diabetes incidence and prevalence in Pima Indians: a 19-fold greater incidence than in Rochester, Minn. *Am J Epidemiol* 108:497–504, 1978
3. Bennett PH, Burch TA, Miller M: Diabetes mellitus in American (Pima) Indians. *Lancet* 2:125–28, 1971
4. Bennett PH, Rushforth NB, Miller M, LeCompte PM: Epidemiologic studies of diabetes in the Pima Indians. In *Recent Progress in Hormone Research*. Greep RO, Ed. New York, Academic Press, 1976, p. 333–76
5. Knowler WC, Pettitt DJ, Saad MF, Bennett PH: Diabetes mellitus in Pima Indians: incidence, risk factors and pathogenesis. *Diabetes Metab Rev* 6:1–27, 1990
6. Haury EM: *The Hohokam: Desert Farmers and Craftsmen, Excavations at Snaketown, 1964–1965*. Tucson, University of Arizona Press, 1976
7. Matson GA, Burch TA, Polesky HF, Swanson J, Sutton HE, Robinson A: Distribution of hereditary factors in the blood of Indians of the Gila River, Arizona. *Am J Phys Anthropol* 29:311–37, 1968
8. Turner CG II: Dental evidence for the peopling of the Americas. In *Early Man in the New World*. Shutler R Jr, Ed. Beverly Hills, CA, Sage Publications, 1983, p. 147–57
9. Williams RC, Steinberg AG, Gershowitz H, Bennett PH, Knowler WC, Pettitt DJ, Butler W, Baird HR, Burch TA, Morse HG, Smith CG: Gm allotypes in Native Americans: evidence for three distinct migrations across the Bering land bridge. *Am J Phys Anthropol* 66:1–19, 1985
10. Wallace DC, Garrison K, Knowler WC: Dramatic founder effects in Amerindian mitochondrial DNAs. *Am J Phys Anthropol* 68:149–55, 1985
11. Russell F: *The Pima Indians. Twenty-sixth Annual Report of the Bureau of American Ethnology*, Washington, DC, 1904–1905
12. Castetter EF, Bell WH: Pima and Papago Indian Agriculture. In *Inter-Americana Studies I*. Albuquerque, The University of New Mexico Press, 1942, p. 1–244
13. Hrdlička A: Physiological and medical observations among the Indians of southwestern United States and northern Mexico. Smithsonian Institution, Bureau of American Ethnology, *Bulletin* 24, Washington, DC, 1908
14. Joslin EP: The universality of diabetes. *JAMA* 115:2033–38, 1940
15. Cohen BM: Diabetes mellitus among Indians of the American Southwest: its prevalence and clinical characteristics in a hospitalized population. *Ann Intern Med* 40:588–99, 1954
16. Parks JH, Waskow E: Diabetes among the Pima Indians of Arizona. *Ariz Med* 18:99–106, 1961
17. Bennett PH, Knowler WC: Increasing prevalence of diabetes in the Pima (American) Indians over a ten-year period. In *Proc. 10th Congress of the IDF*. Waldhausl WK, Ed. Amsterdam, Excerpta Med., 1979, p. 507–11
18. Knowler WC, Pettitt DJ, Savage PJ, Bennett PH: Diabetes incidence in Pima Indians: contributions of obesity and parental diabetes. *Am J Epidemiol* 113:144–56, 1981
19. Savage PJ, Bennett PH, Senter RG, Miller M: High prevalence of diabetes in young Pima Indians. *Diabetes* 28:937–42, 1979
20. Katzef HL, Savage PJ, Barclay-White B, Nagulesparan M, Bennett PH: C-peptide measurement in the differentiation of type 1 (insulin-dependent) and type 2 (non-insulin-dependent) diabetes mellitus. *Diabetologia* 28:264–68, 1985
21. Knowler WC, Bennett PH, Bottazzo GF, Doniach D: Islet cell antibodies and diabetes mellitus in Pima Indians. *Diabetologia* 17:161–64, 1979
22. Menard L, Dempsey ME, Blankstein LA, Aleyassine H, Wacks M, Soeldner JS: Quantitative determination of glycosylated hemoglobin A₁ by agar gel electrophoresis. *Clin Chem* 26:1598–1602, 1980
23. Kaplan EB, Elston RC: *A Subroutine Package for Maximum Likelihood Estimation*. The Institute of Statistics, The University of North Carolina 1972, p. I–IX (Inst. Statis. Mimeo Ser., no. 823)
24. Rushforth NB, Bennett PH, Steinberg AG, Burch TA, Miller M: Diabetes in the Pima Indians: evidence of bimodality in glucose tolerance distributions. *Diabetes* 20:756–65, 1971
25. Dorf A, Ballantine EJ, Bennett PH, Miller M: Retinopathy in Pima Indians: relationships to glucose level, duration of diabetes, age at diagnosis of diabetes, and age at examination in a population with a high prevalence of diabetes mellitus. *Diabetes* 25:554–60, 1976

26. Rushforth NB, Miller M, Bennett PH: Fasting and two-hour post-load glucose levels for the diagnosis of diabetes: the relationship between glucose levels and complications of diabetes in the Pima Indians. *Diabetologia* 16:373-79, 1979
27. Flock EV, Bennett PH, Savage PJ, Webner CJ, Howard BV, Rushforth NB, Miller M: Bimodality of glycosylated hemoglobin distribution in Pima Indians: relationship to fasting hyperglycemia. *Diabetes* 28:984-89, 1979
28. Pettitt DJ, Knowler WC, Lisse JR, Bennett PH: Development of retinopathy and proteinuria in relation to plasma-glucose concentrations in Pima Indians. *Lancet* 2:1050-52, 1980
29. Al Sayegh H, Jarrett RJ: Oral glucose-tolerance tests and the diagnosis of diabetes: results of a prospective study based on the Whitehall Survey. *Lancet* 2:431-33, 1979
30. Jarrett RJ, Keen H: Hyperglycemia and diabetes mellitus. *Lancet* 2:1009-12, 1976
31. National Diabetes Data Group: Classification and diagnosis of diabetes mellitus and other categories of glucose intolerance. *Diabetes* 28:1039-57, 1979
32. World Health Organization: *WHO Expert Committee on Diabetes Mellitus. Second Report*. Geneva, World Health Org., 1980, 9-14 (Tech. Rep. Ser., no. 646)
33. World Health Organization: *Diabetes Mellitus: Report of a WHO Study Group*. Geneva, World Health Org., 1985, 9-17 (Tech. Rep. Ser., no. 727)
34. Harris MI, Hadden WC, Knowler WC, Bennett PH: International criteria for the diagnosis of diabetes and impaired glucose tolerance. *Diabetes Care* 8:562-67, 1985
35. Pettitt DJ, Lisse JR, Knowler WC, Bennett PH: Mortality as a function of obesity and diabetes mellitus. *Am J Epidemiol* 115: 359-66, 1982
36. Knowler WC, Pettitt DJ, Lillioja S, Nelson RG: Genetic and environmental factors in the development of diabetes mellitus in Pima Indians. In *Genetic Susceptibility to Environmental Factors—A Challenge for Public Intervention*. Smith U, Eriksson S, Lindgärde F, Eds. Stockholm, Almqvist & Wiksell International, 1988, p. 67-74
37. Prior AM, Beaglehole R, Davidson F, Salmond CE: The relationship of diabetes, blood lipids, and uric acid levels in Polynesians. *Adv Metab Disord* 9:241-60, 1978
38. Zimmet P, Taft P, Guinea A, Guthrie W, Thoma K: The high prevalence of diabetes mellitus on a Central Pacific Island. *Diabetologia* 13:111-15, 1977
39. Zimmet P, Faaiuso S, Ainuu J, Whitehouse S, Milne B, DeBoer W: The prevalence of diabetes in the rural and urban Polynesian population of Western Samoa. *Diabetes* 30:45-51, 1981
40. Cohen AM: Prevalence of diabetes among different ethnic Jewish groups in Israel. *Metabolism* 10:50-58, 1961
41. Kawate R, Yamakido M, Nishimoto Y, Bennett PH, Hamman RF, Knowler WC: Diabetes mellitus and its complications in Japanese migrants on the island of Hawaii. *Diabetes Care* 2:161-70, 1979
42. Taylor R, Zimmet P: Migrant studies in diabetes epidemiology. In *Diabetes in Epidemiological Perspective*. Mann JI, Pyörälä K, Teuscher A, Eds. New York, Churchill Livingstone, Inc., 1983, p. 58-77
43. West KM: Diabetes in American Indians and other native populations of the New World. *Diabetes* 23:841-55, 1974
44. Knowler WC, Savage PJ, Nagulesparan M, Howard BV, Pettitt DJ, Lisse JR, Aronoff SL, Bennett PH: Obesity, insulin resistance, and diabetes mellitus in the Pima Indians. In *The Genetics of Diabetes Mellitus*. Köbberling J, Tattersall R, Eds. Sero Symposium No. 47. London, Academic Press, 1982, p. 243-50
45. Knowler WC, Pettitt DJ, Bennett PH, Williams RC: Diabetes mellitus in the Pima Indians: genetic and evolutionary considerations. *Am J Phys Anthropol* 62: 107-14, 1983
46. Pettitt DJ, Baird HR, Aleck KA, Bennett PH, Knowler WC: Excessive obesity in offspring of Pima Indian women with diabetes during pregnancy. *N Engl J Med* 308:242-45, 1983
47. Pettitt DJ, Knowler WC, Bennett PH, Aleck KA, Baird HR: Obesity in offspring of diabetic Pima Indians women despite normal birth weight. *Diabetes Care* 10: 76-80, 1987
48. Pettitt DJ, Aleck KA, Baird HR, Carraher MJ, Bennett PH, Knowler WC: Congenital susceptibility to NIDDM: role of intrauterine environment. *Diabetes* 37: 622-28, 1988
49. Pettitt DJ, Knowler WC: Diabetes and obesity in the Pima Indians: a cross-generational vicious cycle. *J Obesity and Weight Regulation* 7:61-75, 1988
50. Knowler WC, Williams RC, Pettitt DJ, Steinberg AG: $Gm^{3,5,13,14}$ and type 2 diabetes mellitus: an association in American Indians with genetic admixture. *Am J Hum Genet* 43:520-26, 1988
51. Williams RC, Knowler WC, Butler WJ, Pettitt DJ, Lisse JR, Bennett PH, Mann DI, Johnson AH, Terasaki PI: HLA-A2 and type 2 (insulin independent) diabetes mellitus in Pima Indians: an association of allele frequency with age. *Diabetologia* 21:460-63, 1981
52. Knowler WC, Pettitt DJ, Vasquez B, Rotwein PS, Andreone T, Permutt MA: Polymorphism in the 5' flanking region of the human insulin gene: relationships with non-insulin-dependent diabetes mellitus, glucose and insulin concentrations, and diabetes treatment in the Pima Indians. *J Clin Invest* 74:2129-35, 1984
53. Elbein SC, Corsetti L, Ullrich A, Permutt MA: Multiple restriction fragment length polymorphisms at the insulin receptor locus: a highly informative marker for linkage analysis. *Proc Natl Acad Sci USA* 83:5223-27, 1986
54. Knowler WC: The genetics of type 2 diabetes. In *Diabetes 1988*. Larkins RG, Zimmet PZ, Chisholm DJ, Eds. New York, Excerpta Med., Elsevier Science Publishers, 1989, p. 489-92
55. Bogardus C, Lillioja S, Nyomba G, Zurlo F, Swinburn B, Esposito-DelPuente A, Knowler WC, Ravussin E, Mott DM, Bennett PH: Distribution of in vivo insulin action in Pima Indians as a mixture of three normal distributions. *Diabetes* 38: 1423-32, 1989
56. Reid JM, Fullmer SD, Pettigrew KD, Burch TA, Bennett PH, Miller M, Whedon GD: Nutrient intake of Pima Indian women: relationships to diabetes mellitus and gallbladder disease. *Am J Clin Nutr* 24:1281-89, 1971
57. Bennett PH, Knowler WC, Baird HR,

- Butler WJ, Pettitt DJ, Reid J: Diet and development of non-insulin-dependent diabetes mellitus: an epidemiological perspective. In *Diet, Diabetes and Atherosclerosis*, Pozza G, Micossi P, Catapano AL, Paoletti R, Eds. New York, Raven Press, 1984, p. 109-19
58. Bennett PH, Knowler WC, Pettitt DJ, Carraher MJ, Vasquez B: Longitudinal studies of the development of diabetes in the Pima Indians. In *Advances in Diabetes Epidemiology*. INSERM Symposium No. 22. Eschwege E, Ed. Amsterdam, Elsevier Biomedical Press, 1982, p. 65-74
59. Saad MF, Knowler WC, Pettitt DJ, Nelson RG, Bennett PH: Transient impaired glucose tolerance in Pima Indians: is it important? *Br Med J* 297:1438-41, 1988
60. Saad MF, Knowler WC, Pettitt DJ, Nelson RG, Mott DM, Bennett PH: The natural history of impaired glucose tolerance in the Pima Indians. *N Engl J Med* 319:1500-06, 1988
61. King H, Zimmet P, Raper LA, Balkau B: The natural history of impaired glucose tolerance in the Micronesian population of Nauru: a six-year follow-up study. *Diabetologia* 26:39-43, 1984
62. Sicree RA, Zimmet Z, King OM, Coventry JS: Plasma insulin response among Nauruans: prediction of deterioration in glucose tolerance over 6 years. *Diabetes* 36:179-86, 1987
63. Jarrett RJ, Keen H, Fuller JH, McCartney P: Worsening to diabetes in men with impaired glucose tolerance (borderline diabetes). *Diabetologia* 16:25-30, 1979
64. Sartor G, Scherstén B, Carlström S, Melander A, Norden Å, Persson G: Ten-year follow-up of subjects with impaired glucose tolerance: prevention of diabetes by tolbutamide and diet regulation. *Diabetes* 29:41-49, 1980
65. Keen H, Jarrett RJ, McCartney P: The ten-year follow-up of the Bedford Survey (1962-1972): glucose tolerance and diabetes. *Diabetologia* 22:73-78, 1982
66. Sasaki A, Suzuki T, Horiuchi N: Development of diabetes in Japanese subjects with impaired glucose tolerance: a seven-year follow-up study. *Diabetologia* 22:154-57, 1982
67. Nagulesparan M, Savage PJ, Mott DM, Johnson GC, Unger RH, Bennett PH: Increased insulin resistance in obese glucose-intolerant Southwestern American Indians: evidence for a defect not explained by obesity. *J Clin Endocrinol Metab* 51:739-43, 1980
68. Lillioja S, Mott DM, Howard BV, Bennett PH, Yki-Jarvinen H, Freymond D, Nyomba BL, Zurlø F, Swinburn B, Bogardus C: Impaired glucose tolerance as a disorder of insulin action: longitudinal and cross-sectional studies in Pima Indians. *N Engl J Med* 318:1217-25, 1988
69. Herbert V, Lau K, Gottlieb CW, Bleicher SJ: Coated charcoal immunoassay of insulin. *J Clin Endocrinol Metab* 25:1375-84, 1965
70. Kaplan EL, Meier P: Nonparametric estimation from incomplete observations. *J Am Statist Assoc* 53:457-81, 1958
71. Cox DR: Regression models and life tables (with discussion). *J R Statist Soc B* 34:187-220, 1972
72. Harrell FE Jr: The PHGLM procedure. In *SUGI Supplemental Library User's Guide*. Version 5 Ed., Chap. 34, Cary, NC, SAS Institute, 1986, p. 437-65
73. Savage PJ, Dippe SE, Bennett PH, Gordon P, Miller M: Hyperinsulinemia and hypoinsulinemia. Insulin responses to oral carbohydrate over a wide spectrum of glucose tolerance. *Diabetes* 24:362-68, 1975
74. Bogardus C, Lillioja S, Howard BV, Reaven G, Mott D: Relationships between insulin secretion, insulin action, and fasting plasma glucose concentrations in nondiabetic and non-insulin-dependent diabetic subjects. *J Clin Invest* 74:1238-46, 1984
75. Reaven G, Miller R: Study of the relationship between glucose and insulin responses to an oral glucose load in man. *Diabetes* 17:560-69, 1968
76. Welborn TA, Stenhouse NS, Johnston CG: Factors determining serum insulin response in a population sample. *Diabetologia* 5:263-66, 1969
77. Zimmet P, Whitehouse S, Alford F, Chisholm D: The relationship of insulin response to a glucose stimulus over a wide range of glucose tolerance. *Diabetologia* 15:23-27, 1978
78. Martin FIR, Wyatt GB, Griew AR, Haurahelia M, Higginbotham L: Diabetes mellitus in urban and rural communities in Papua, New Guinea: studies of prevalence and plasma insulin. *Diabetologia* 18:369-74, 1980
79. Zimmet P, Whitehouse S, Kiss J: Ethnic variability in the plasma insulin response to oral glucose in Polynesian and Micronesian subjects. *Diabetes* 28:624-28, 1979
80. DeFronzo RA: The triumvirate: B-cell, muscle, liver. A collusion responsible for NIDDM. *Diabetes* 37:667-87, 1988
81. Saad MF, Knowler WC, Pettitt DJ, Nelson RG, Mott DM, Bennett PH: Sequential changes in serum insulin concentration during development of non-insulin-dependent diabetes. *Lancet* 1:1356-59, 1989
82. Kolterman OG, Insel J, Saekow M, Olefsky JM: Mechanisms of insulin resistance in human obesity: evidence for receptor and post-receptor defects. *J Clin Invest* 65:1272-84, 1980
83. Kashiwagi A, Bogardus C, Lillioja S, Huecksteadt TP, Brady D, Verso MA, Foley JE: In vitro insensitivity of glucose transport and antilipolysis to insulin due to receptor and post-receptor abnormalities in obese Pima Indians with normal glucose tolerance. *Metabolism* 33:772-77, 1984
84. Lillioja S, Young AA, Cutler CL, Ivy JL, Abbott WGH, Zawadzki JK, Yki-Jarvinen HJ, Christin L, Secomb TW, Bogardus C: Skeletal muscle capillary density and fiber type are possible determinants of in vivo insulin resistance in man. *J Clin Invest* 80:415-24, 1987
85. Fink RI, Kolterman OG, Griffin J, Olefsky JF: Mechanisms of insulin resistance of aging. *J Clin Invest* 7:1523-35, 1983
86. Rowe JW, Minaker KL, Pallota JA, Flier JS: Characterization of the insulin resistance of aging. *J Clin Invest* 71:1581-87, 1983
87. Yki-Jarvinen H, Koivisto VA: Effects of body composition on insulin sensitivity. *Diabetes* 32:965-69, 1983
88. Saad MF, Knowler WC, Pettitt DJ, Nelson RG, Charles MA, Bennett PH: A two-step model for development of non-insulin-dependent diabetes. *Am J Med* 90:229-35, 1991
89. Lillioja S, Mott DM, Zawadzki JK, Young

- AA, Abbott WGH, Knowler WC, Bennett PH, Moll P, Bogardus C: In vivo insulin action is familial characteristic in non-diabetic Pima Indians. *Diabetes* 36: 1329-35, 1987
90. Bogardus C, Lillioja S, Foley J, Christin L, Freymond D, Nyomba B, Bennett PH, Reaven G, Salans L: Insulin resistance predicts the development of non-insulin-dependent diabetes mellitus in Pima Indians (Abstract). *Diabetes* 36:47A, 1987
91. Schraer CD, Lanier AP, Boyko EJ, Gohdes D, Murphy NJ: Prevalence of diabetes in Alaskan Eskimos, Indians and Aleuts. *Diabetes Care* 11:693-700, 1988
92. Gardner LI Jr, Stern MP, Haffner SM, Gaskill SP, Hazuda HP, Relethford JH, Eifler CW: Prevalence of diabetes in Mexican-Americans: relationship to percent of gene pool derived from Native American sources. *Diabetes* 33:86-92, 1984